Experiment 13: OFDM Channel Estimation

Using a Sionna RT-Derived Multipath Channel at 5.9 GHz

V2X–ISAC Research
Sionna RT Simulation Environment

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Abstract

This experiment validates an Orthogonal Frequency Division Multiplexing (OFDM) channel-estimation pipeline using a discrete multipath channel derived from ray tracing in NVIDIA Sionna RT.

The simulated transmitter and receiver operate at a carrier frequency of 5.9000 GHz, with a sampling rate of 1.0000×10^2 MHz and an FFT size of 1024. Twelve valid propagation paths are obtained from the ray-tracing simulation and represented by a finite-length discrete-time complex channel.

A Quadrature Phase Shift Keying (QPSK) OFDM symbol is transmitted through this channel. At the receiver, the cyclic prefix is removed, an FFT is applied, and the channel frequency response is estimated using the known transmitted symbols.

The measured relative complex Root Mean Square Error (RMSE) between the estimated and reference frequency responses is $3.81651769 \times 10^{-16}$, with noise disabled. The estimated and reference channel responses overlap closely, and the phase and normalized-error plots show residuals consistent with floating-point precision.

These results demonstrate numerical consistency between the OFDM processing chain and the discrete channel used to generate the reference response. They do not, by themselves, establish the physical accuracy of the propagation model or validate real-world V2X sensing performance.

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1 Introduction

Orthogonal Frequency Division Multiplexing is widely used in modern wireless communication systems because it divides a wideband channel into multiple orthogonal subcarriers.

Under appropriate synchronization and cyclic-prefix conditions, a frequency-selective multipath channel can be represented as a complex multiplicative coefficient on each subcarrier.

In this experiment, a ray-tracing-derived channel is obtained using Sionna RT. The channel is then used in a baseband OFDM simulation to validate the channel-estimation process.

The objectives are:

1. Obtain propagation paths from a Sionna RT scene.
2. Construct a discrete-time complex multipath channel.
3. Generate and transmit a QPSK OFDM symbol through the discrete channel.
4. Estimate the channel frequency response.
5. Compare the estimated response against a reference response calculated from the same discrete channel.

This is a signal-processing validation experiment and an intermediate step toward more realistic V2X Integrated Sensing and Communication (ISAC) experiments.

2 Simulation Environment

The simulation uses the built-in Sionna RT `simple_street_canyon` scene.

The Mitsuba rendering backend is configured as `llvm_ad_mono_polarized`, allowing the ray-tracing calculation to run using the LLVM backend.

2.1 Transmitter and receiver geometry

The transmitter and receiver positions are

$$\mathbf{p}_{\mathrm{TX}} = \begin{bmatrix} -20 & 0 & 2 \end{bmatrix}^T \ \mathrm{m}, \tag{1}$$

$$\mathbf{p}_{\mathrm{RX}} = \begin{bmatrix} 20 & 0 & 2 \end{bmatrix}^T \ \mathrm{m}. \tag{2}$$

Their Euclidean separation is

$$\begin{aligned} d_{\mathrm{TX,RX}} &= \|\mathbf{p}_{\mathrm{RX}} - \mathbf{p}_{\mathrm{TX}}\|_2 && (3)\\ &= \sqrt{(20-(-20))^2 + (0-0)^2 + (2-2)^2} && (4)\\ &= 4.0000 \times 10^1 \,\mathrm{m}. && (5) \end{aligned}$$

The direct-path propagation delay, assuming a clear line-of-sight path, is

$$\tau_{\mathrm{LOS}} = \frac{d_{\mathrm{TX,RX}}}{c}, \tag{6}$$

where $c = 299\,792\,458\ \mathrm{m\,s^{-1}}$ is the speed of light.

Therefore,

$$\tau_{\mathrm{LOS}} \approx 1.3343 \times 10^{2}\,\mathrm{ns}. \tag{7}$$

This agrees with the earliest path delay reported by the ray-tracing simulation.

2.2 Carrier frequency and wavelength

The carrier frequency is

$$f_c = 5.9000\,\mathrm{GHz}. \tag{8}$$

The corresponding free-space wavelength is

$$\lambda_c = \frac{c}{f_c}. \tag{9}$$

Hence,

$$\lambda_c \approx 5.0800 \times 10^{-2}\,\mathrm{m} = 5.0800\,\mathrm{cm}. \tag{10}$$

The wavelength is relevant to propagation and antenna design. The present baseband experiment does not directly simulate a physical RF front end.

3 Ray-Tracing Channel Model

3.1 Continuous-time multipath representation

A time-invariant multipath channel can be represented as

$$h_c(t) = \sum_{\ell=0}^{L-1} a_\ell \delta(t - \tau_\ell), \tag{11}$$

where L is the number of valid propagation paths, a_ℓ is the complex coefficient of path ℓ, τ_ℓ is its propagation delay, and $\delta(t)$ is the Dirac delta function.

The complex path coefficient incorporates the amplitude and phase represented by the ray-tracing model.

The simulation reported

$$L = 12 \tag{12}$$

valid paths, with approximate delay limits

$$\tau_{\min} = 1.3343 \times 10^{2}\,\mathrm{ns}, \tag{13}$$

$$\tau_{\max} = 2.8328 \times 10^{2}\,\mathrm{ns}. \tag{14}$$

The delay span is therefore

$$\Delta\tau_{\text{paths}} = \tau_{\max} - \tau_{\min} \tag{15}$$
$$\approx 1.4986 \times 10^{2}\,\text{ns}. \tag{16}$$

This is the span between the earliest and latest reported rays. It is not necessarily the RMS delay spread, which requires a power-weighted calculation.

3.2 Discrete-time channel construction

The sampling rate is

$$f_s = 1.0000 \times 10^{2}\,\text{MHz}. \tag{17}$$

The sampling interval is

$$T_s = \frac{1}{f_s} = 1.0000 \times 10^{1}\,\text{ns}. \tag{18}$$

The fractional sample index for path ℓ is

$$\nu_\ell = \frac{\tau_\ell}{T_s} = \tau_\ell f_s. \tag{19}$$

Because ν_ℓ need not be an integer, directly rounding each ray delay to a sample can introduce delay quantization error.

The implementation approximates each fractional-delay contribution using a windowed-sinc interpolation kernel.

For a fractional sample position ν_ℓ, the kernel is

$$w_\ell[n] = \text{sinc}(n - \nu_\ell)\,\text{sinc}\left(\frac{n - \nu_\ell}{K+1}\right), \tag{20}$$

over the retained finite support, where $K = 16$ and

$$\text{sinc}(x) = \begin{cases} \dfrac{\sin(\pi x)}{\pi x}, & x \neq 0, \\ 1, & x = 0. \end{cases} \tag{21}$$

The discrete channel is approximated by

$$h[n] = \sum_{\ell=0}^{L-1} a_\ell w_\ell[n]. \tag{22}$$

Contributions outside the allocated sample window are discarded.

The maximum delay window is

$$T_{\max} = 6.0000 \times 10^{2}\,\text{ns}. \tag{23}$$

The allocated number of samples is

$$N_h = \left\lceil \frac{T_{\max}}{T_s} \right\rceil \tag{24}$$

$$= \left\lceil \frac{600}{10} \right\rceil \tag{25}$$

$$= 60. \tag{26}$$

Thus, the discrete channel contains 60 samples and has a nominal memory of $N_h - 1 = 59$ samples.

The fractional-delay interpolation is an approximation. Its finite support and boundary truncation can affect the sampled channel, especially near the allocated window boundaries.

4 OFDM System Configuration

Table 1 summarizes the simulation parameters.

Table 1: OFDM simulation parameters

Parameter	**Value**
Carrier frequency	5.9000 GHz
Sampling rate	1.0000×10^2 MHz
FFT size N	1024
Modulation	QPSK
Cyclic-prefix length	128 samples
Cyclic-prefix duration	1.2800 µs
Useful symbol duration	1.0240×10^1 µs
Subcarrier spacing	9.7656×10^1 kHz
Discrete channel length	60 samples
Valid ray paths	12
Noise	Disabled
Random seed	2026

4.1 Subcarrier spacing

The subcarrier spacing is

$$\Delta f = \frac{f_s}{N}. \tag{27}$$

Substitution gives

$$\Delta f = \frac{100 \times 10^6}{1024} \tag{28}$$

$$= 9.7656 \times 10^4 \,\text{Hz} \tag{29}$$

$$= 9.7656 \times 10^1 \,\text{kHz}. \tag{30}$$

4.2 Useful symbol duration

The useful OFDM symbol duration is

$$T_u = \frac{N}{f_s} = \frac{1}{\Delta f}. \tag{31}$$

Therefore,

$$T_u = \frac{1024}{100 \times 10^6} = 1.0240 \times 10^1\,\mu\text{s}. \tag{32}$$

4.3 Cyclic-prefix duration

The cyclic prefix contains 128 samples. Its duration is

$$T_{\text{CP}} = \frac{N_{\text{CP}}}{f_s}. \tag{33}$$

Thus,

$$T_{\text{CP}} = \frac{128}{100 \times 10^6} = 1.2800\,\mu\text{s}. \tag{34}$$

The total OFDM symbol duration, excluding any additional framing or guard intervals, is

$$\begin{aligned} T_{\text{sym}} &= T_u + T_{\text{CP}} && (35)\\ &= 1.0240 \times 10^1\,\mu\text{s} + 1.2800\,\mu\text{s} && (36)\\ &= 1.1520 \times 10^1\,\mu\text{s}. && (37) \end{aligned}$$

5 QPSK OFDM Signal Generation

5.1 QPSK mapping

Independent bits $b_I[k], b_Q[k] \in \{0, 1\}$ are generated for each subcarrier.

The QPSK symbol is

$$X[k] = \frac{(1 - 2b_I[k]) + j(1 - 2b_Q[k])}{\sqrt{2}}, \tag{38}$$

where $j = \sqrt{-1}$.

The symbol magnitude satisfies

$$\begin{aligned} |X[k]|^2 &= \frac{(1 - 2b_I[k])^2 + (1 - 2b_Q[k])^2}{2} && (39)\\ &= \frac{1+1}{2} && (40)\\ &= 1. && (41) \end{aligned}$$

Consequently,

$$|X[k]| = 1 \tag{42}$$

for all subcarriers, and the transmitted symbols are non-zero.

5.2 OFDM modulation

The useful time-domain OFDM symbol is generated using the inverse discrete Fourier transform:

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j2\pi kn/N}, \quad 0 \le n < N. \tag{43}$$

The cyclic prefix is formed by copying the last N_{CP} samples to the beginning of the symbol:

$$x_{\mathrm{CP}}[n] = \begin{cases} x[n+N], & -N_{\mathrm{CP}} \le n < 0, \\ x[n], & 0 \le n < N. \end{cases} \tag{44}$$

The transmitted symbol therefore contains the cyclic prefix followed by the useful OFDM samples.

6 Channel Propagation and OFDM Reception

6.1 Discrete-time channel model

The received signal is modelled as the linear convolution

$$y[n] = \sum_{m=0}^{N_h-1} h[m] x_{\mathrm{CP}}[n-m]. \tag{45}$$

Additive noise is disabled in this experiment. A more general model would include a noise term $z[n]$:

$$y[n] = \sum_{m=0}^{N_h-1} h[m] x_{\mathrm{CP}}[n-m] + z[n]. \tag{46}$$

6.2 Cyclic-prefix condition

For a discrete channel of length N_h, the channel memory is $N_h - 1$ samples. The usual cyclic-prefix condition is

$$N_{\mathrm{CP}} \ge N_h - 1. \tag{47}$$

Here,

$$128 \ge 60 - 1 = 59. \tag{48}$$

Thus, the cyclic prefix is sufficiently long for the constructed discrete channel, assuming correct timing and a channel that remains static during the OFDM symbol.

After cyclic-prefix removal, the useful received symbol can be represented as a circular convolution. The DFT converts circular convolution into multiplication:

$$Y[k] = H[k]X[k]. \tag{49}$$

This relationship assumes a sufficiently long cyclic prefix, correct timing, and no carrier-frequency offset.

7 Reference Channel and Channel Estimation

7.1 Reference frequency response

The reference frequency response is calculated by taking the N-point DFT of the discrete channel:

$$H_{\text{ref}}[k] = \sum_{n=0}^{N_h-1} h[n]e^{-j2\pi kn/N}. \tag{50}$$

In the implementation, the 60-sample channel is zero-padded to length 1024 before its FFT is calculated.

7.2 Least-squares channel estimation

The frequency-domain input-output relationship is

$$Y[k] = H[k]X[k]. \tag{51}$$

Since the transmitted QPSK symbol is known, the channel estimate is

$$\widehat{H}[k] = \frac{Y[k]}{X[k]}. \tag{52}$$

This is the single-observation least-squares estimate, which minimizes

$$\widehat{H}[k] = \arg\min_{g\in\mathbb{C}} \left|Y[k] - gX[k]\right|^2. \tag{53}$$

Since the QPSK symbols are non-zero, the estimate is well-defined for every subcarrier.

8 Mathematical Proof and Error Metrics

8.1 Proof of ideal channel-estimation recovery

Under the static-channel, correct-timing, sufficient-cyclic-prefix, and noise-free assumptions,

$$Y[k] = H_{\text{ref}}[k]X[k]. \tag{54}$$

Using the estimator,

$$\widehat{H}[k] = \frac{Y[k]}{X[k]} \tag{55}$$

$$= \frac{H_{\text{ref}}[k]X[k]}{X[k]} \tag{56}$$

$$= H_{\text{ref}}[k]. \tag{57}$$

Therefore,

$$\boxed{\widehat{H}[k] = H_{\text{ref}}[k]} \tag{58}$$

for every subcarrier under the stated assumptions.
Define the estimation error as

$$e[k] = \widehat{H}[k] - H_{\text{ref}}[k]. \tag{59}$$

Equation (58) implies

$$e[k] = 0 \tag{60}$$

in exact arithmetic. Thus, the ideal error power is

$$\frac{1}{N}\sum_{k=0}^{N-1} |e[k]|^2 = 0. \tag{61}$$

A numerical implementation uses finite-precision arithmetic, so its measured residual is not necessarily exactly zero.

8.2 Relative complex RMSE

The error RMS is

$$E_{\text{RMS}} = \sqrt{\frac{1}{N}\sum_{k=0}^{N-1} |\widehat{H}[k] - H_{\text{ref}}[k]|^2}. \tag{62}$$

The reference RMS is

$$H_{\text{RMS}} = \sqrt{\frac{1}{N}\sum_{k=0}^{N-1} |H_{\text{ref}}[k]|^2}. \tag{63}$$

The relative complex RMSE is

$$\text{RMSE}_{\text{rel}} = \frac{E_{\text{RMS}}}{H_{\text{RMS}}}. \tag{64}$$

The measured values were

$$H_{\text{RMS}} = 1.135065 \times 10^{-4}, \tag{65}$$

$$E_{\text{RMS}} = 4.331997 \times 10^{-20}. \tag{66}$$

Consequently,

$$\mathrm{RMSE}_{\mathrm{rel}} = \frac{4.331997 \times 10^{-20}}{1.135065 \times 10^{-4}} \tag{67}$$

$$\approx 3.81651769 \times 10^{-16}. \tag{68}$$

As a percentage,

$$\boxed{\mathrm{RMSE}_{\mathrm{rel}}(\%) \approx 3.81651769 \times 10^{-14}\%} \tag{69}$$

The program prints the percentage to six decimal places, so it appears as 0.000000%. This rounded display does not mean that the measured residual is exactly zero.

8.3 Phase error

The phase difference is calculated as

$$\Delta\phi[k] = \arg\left(\widehat{H}[k]H^*_{\mathrm{ref}}[k]\right), \tag{70}$$

where $H^*_{\mathrm{ref}}[k]$ denotes the complex conjugate of the reference response.

Subcarriers near deep spectral nulls are excluded from the phase RMSE calculation because the phase of a near-zero complex value is numerically sensitive.

The reported phase RMSE away from deep nulls was approximately zero degrees at the program's displayed precision.

9 Experimental Results and Graphical Analysis

9.1 Numerical results

Table 2 summarizes the measured results.

Table 2: Measured results for Experiment 13

Metric	**Result**
Valid ray paths	12
Discrete channel length	60 samples
Reference channel RMS	1.135065×10^{-4}
Estimation error RMS	4.331997×10^{-20}
Relative complex RMSE	$3.81651769 \times 10^{-16}$
Relative complex RMSE (rounded)	0.000000%
Phase RMSE away from deep nulls	Approximately 0°
Noise	Disabled

9.2 Graphical results

Figure 1 shows the channel magnitude comparison, phase error, and normalized absolute error produced by the Python experiment.

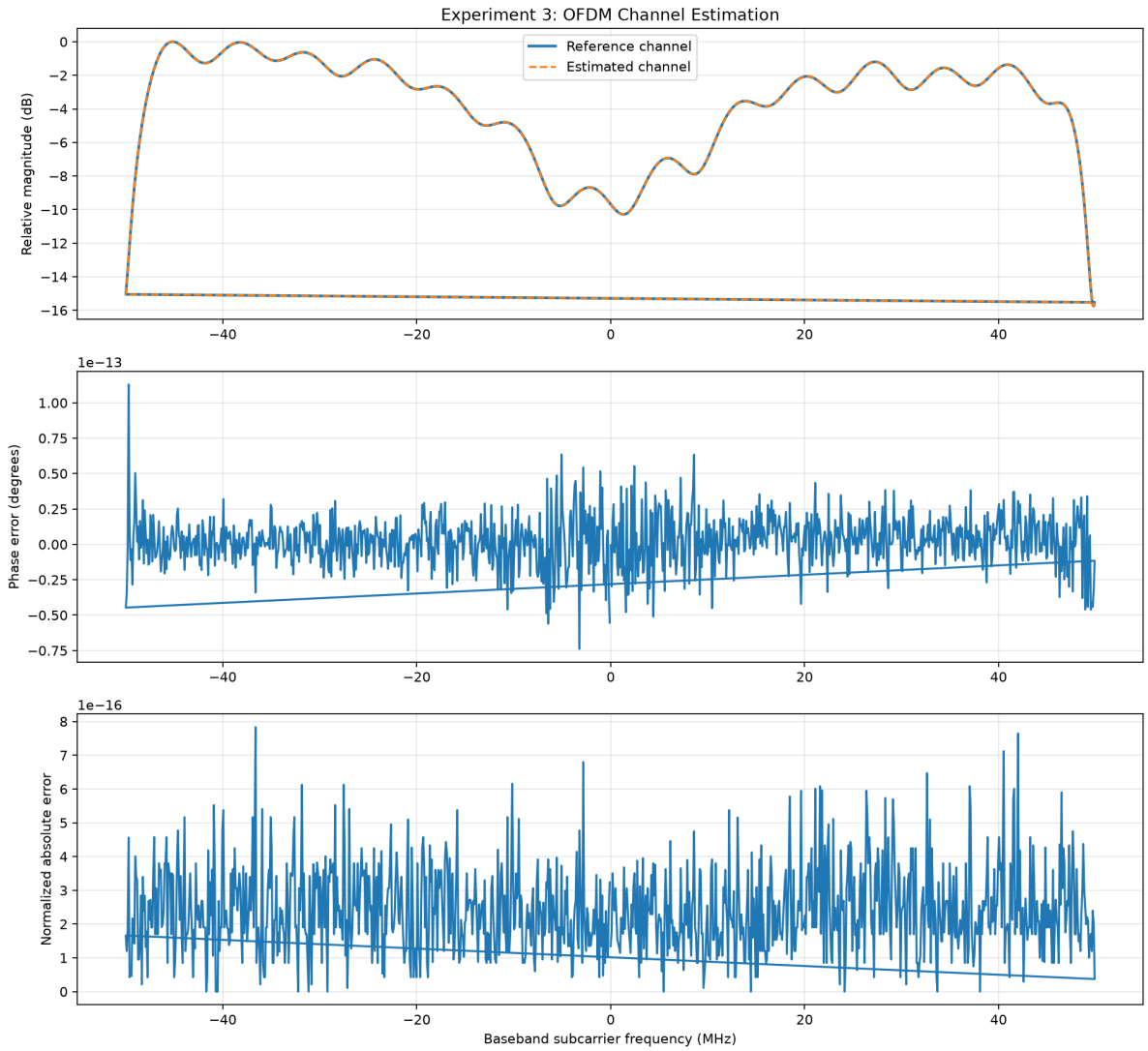


Figure 1: Experiment 13 OFDM channel-estimation results. Top: reference and estimated channel magnitudes in dB. Middle: phase error in degrees. Bottom: normalized absolute complex estimation error.

9.3 Magnitude response: description

The upper panel displays the relative channel magnitude over baseband frequencies from approximately -5.0000×10^{1} MHz to $+5.0000 \times 10^{1}$ MHz.

The solid blue curve represents the reference channel, and the orange dashed curve represents the estimated channel. The two curves overlap closely over the plotted bandwidth.

The channel magnitude varies with frequency, with multiple local maxima and minima and a pronounced attenuation region around the central part of the band. This behaviour is consistent with a frequency-selective multipath channel.

The frequency response of the continuous-time multipath model is

$$H(f) = \sum_{\ell=0}^{L-1} a_\ell e^{-j2\pi f \tau_\ell}. \tag{71}$$

Each propagation path contributes a frequency-dependent phase rotation. The complex contributions can reinforce one another at some frequencies and cancel at others, creating constructive and destructive interference.

The close overlap of the two curves indicates that the estimated response reproduces the frequency-selective shape of the reference discrete channel.

The decibel display also makes small absolute differences difficult to distinguish near deep attenuation regions. Therefore, the magnitude plot should be interpreted alongside the numerical error metrics.

9.4 Phase error: description

The middle panel shows the phase difference between the estimated and reference responses. The vertical axis is scaled to approximately 10^{-13} degrees.

The residual fluctuations are extremely small and are consistent with floating-point effects in complex-valued convolution, FFT processing, and division.

They should not be interpreted as realistic phase noise or physical RF phase error. The reported phase RMSE rounds to zero at the program's displayed precision.

Phase is also poorly conditioned near a deep spectral null. A very small change in the complex response can cause a relatively large change in its phase when its magnitude is close to zero. Excluding deep nulls from the phase RMSE is therefore appropriate.

9.5 Normalized absolute error: description

The lower panel displays the normalized absolute complex error. For each subcarrier, the absolute error is

$$e[k] = \widehat{H}[k] - H_{\mathrm{ref}}[k]. \tag{72}$$

The normalized absolute error is

$$\epsilon[k] = \frac{|\widehat{H}[k] - H_{\mathrm{ref}}[k]|}{H_{\mathrm{RMS}}}, \tag{73}$$

where H_{RMS} is the RMS magnitude of the reference response.

The plot uses a vertical scale of approximately 10^{-16}. This is consistent with the very small relative complex RMSE reported by the program.

The residual variations do not show a substantial frequency-dependent estimation error. Under the noise-free assumptions, they are best interpreted as numerical residuals rather than meaningful physical channel-estimation errors.

9.6 Inference from the combined plots

The three panels provide complementary evidence:

1. The magnitude curves overlap, showing agreement in the frequency-selective channel response.

2. The phase error remains extremely small, showing agreement in the complex phase response away from deep spectral nulls.

3. The normalized absolute error remains at a numerical precision scale, supporting the very small aggregate complex RMSE.

Together, the plots support the conclusion that the channel estimator and the OFDM signal-processing chain are consistent with the discrete channel used to generate the received signal.

10 Overall Inference

The experiment supports the following conclusions.

1. **OFDM processing consistency:** The estimated channel agrees with the reference response calculated from the discrete channel used in the simulation.
2. **Cyclic-prefix adequacy:** The 128-sample cyclic prefix exceeds the 59-sample memory of the constructed channel. It is therefore sufficient under the assumed static-channel and correct-timing conditions.
3. **Multipath frequency selectivity:** The variation of the magnitude response with frequency is consistent with the combined contributions of paths with different delays and complex coefficients.
4. **Numerical accuracy:** The relative complex RMSE of approximately 3.82×10^{-16} indicates near-identical reference and estimated responses in this ideal test.
5. **Scope of validation:** The experiment validates the numerical consistency of the OFDM processing chain. It does not independently validate the physical accuracy of the Sionna RT scene or the performance of a real RF receiver.
6. **Next step:** Noise, synchronization offsets, carrier-frequency offset, and Doppler should be introduced in subsequent experiments to evaluate robustness under less ideal conditions.

11 Limitations

Several limitations must be considered when interpreting the results.

1. **Noise-free operation:** Additive noise and interference are disabled.
2. **Perfect timing:** The receiver extracts the correct OFDM symbol interval.
3. **No carrier-frequency offset:** Frequency synchronization errors and inter-carrier interference are not modelled.
4. **Static channel:** Channel variation and Doppler due to moving vehicles are not included.
5. **Shared reference model:** The estimated response is compared with the FFT of the same discrete channel used to generate the received signal. This is an internal consistency check rather than an independent physical validation.

6. **Fractional-delay approximation:** The finite-support interpolation kernel approximates continuous ray delays and may introduce discretization and boundary effects.

7. **No target estimation:** This experiment does not estimate object range, velocity, or position.

12 Relevance to V2X-ISAC Research

The experiment establishes a baseline OFDM signal-processing chain for subsequent V2X Integrated Sensing and Communication research.

A logical next step is to transform the estimated frequency response into a delay-domain profile and compare detected peaks against known simulated propagation delays.

For monostatic radar, round-trip delay is related to range by

$$R = \frac{c\tau_{\mathrm{rt}}}{2}. \tag{74}$$

The factor of two accounts for propagation from the radar to the target and back to the radar.

The current experiment instead uses spatially separated transmitter and receiver locations. For a scattering point $\mathbf{p}$, the bistatic propagation distance is

$$d_{\mathrm{bistatic}} = \|\mathbf{p} - \mathbf{p}_{\mathrm{TX}}\|_2 + \|\mathbf{p}_{\mathrm{RX}} - \mathbf{p}\|_2. \tag{75}$$

The corresponding propagation delay is

$$\tau_{\mathrm{bistatic}} = \frac{d_{\mathrm{bistatic}}}{c}. \tag{76}$$

Therefore, the monostatic range formula must not be applied indiscriminately to every delay in the current separated transmitter-receiver geometry.

The nominal delay-resolution scale associated with bandwidth B is

$$\Delta\tau \sim \frac{1}{B}. \tag{77}$$

For a conventional monostatic radar, the corresponding nominal range-resolution scale is

$$\Delta R \sim \frac{c}{2B}. \tag{78}$$

For an illustrative bandwidth of $1.0000 \times 10^2\,\mathrm{MHz}$,

$$\Delta R \sim \frac{299\,792\,458}{2(100 \times 10^6)} \tag{79}$$

$$\approx 1.5000\,\mathrm{m}. \tag{80}$$

This is a theoretical bandwidth-based resolution scale, not a measured range-resolution result from Experiment 13. Actual target separability depends on waveform properties, effective bandwidth, SNR, windowing, propagation geometry, and the estimation method.

13 Conclusion

Experiment 13 successfully validated a basic OFDM channel-estimation pipeline using a discrete multipath channel derived from Sionna RT at 5.9000 GHz.

The simulation used a sampling rate of 1.0000×10^2 MHz, an FFT size of 1024, QPSK modulation, and a 128-sample cyclic prefix. The 60-sample discrete channel satisfies the cyclic-prefix memory condition under the assumed static-channel and correct-timing model.

The measured relative complex RMSE was $3.81651769 \times 10^{-16}$. The magnitude, phase-error, and normalized-error plots all support the conclusion that the estimated response reproduces the reference discrete channel to numerical precision.

The result is consistent with the mathematical expectation that a noise-free OFDM receiver with known non-zero transmitted symbols can recover the frequency response of the discrete channel used to generate its received signal.

However, this ideal numerical agreement does not establish the performance of a physical V2X sensing system. Future experiments should evaluate delay-domain estimation and progressively introduce noise, timing offsets, Doppler, and more realistic sensing geometries.

These steps will help determine whether the validated signal-processing chain can support practical V2X ISAC object detection and localization.